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CNN With Large Data Achieves True Zero-Training in Online P300 Brain-Computer Interface

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ABSTRACT Each person has his or her own distinct event-related potential (ERP) signals. Thus, traditional brain-computer interface (BCI) systems require a calibration process in which the subject's data are extracted in order to train machine-learning classifiers. Despite past efforts to eliminate this process, often referred to as “zero-training,” BCI systems' best performance is achievable only with some level of calibration. This tedious process is one of the factors that have limited the use of BCI systems in the real world. Meanwhile, convolutional neural networks (CNN) have been proven to be useful in distinguishing neurophysiological features. In this study, we investigated whether an existing convolutional neural network (CNN) combined with large ERP samples ($n = 99,000$) can achieve zero-training in a P300 BCI speller system. As a result, the zero-trained CNN achieved comparable performance (89%, $p < 0.05$) when compared to traditional approaches (94%) in an offline study. We also demonstrate that the constructed CNN works successfully in an online experiment in which twelve BCI subjects achieved 85% mean accuracy without calibration, compared to 82% (but the difference was not significant, $p > 0.05$) with calibration. Additionally, we illustrate a hybrid approach in order to further enhance performance, which adaptively updates a linear classifier using label information generated from a zero-trained CNN. With this technique, the hybrid approach achieved reasonable performance (92%), showing no statistical difference ($p > 0.05$) when compared to the traditional approach in the same offline data.

INDEX TERMS Brain-computer interface, machine learning, P300 speller, zero-training.

I. INTRODUCTION

Brain-computer interface (BCI) applications have begun to increasingly find their way into our lives through areas such as prosthesis, rehabilitation, and games [1], [2]. A large portion of these applications use event-related potential (ERP) elicited from the brain [3], [4]. They often use visual, tactile, or auditory stimulus in order to elicit ERP signals, as well as hybrid stimulus [5]–[8]. However, most ERP-based BCI applications cannot effectively distinguish between target and non-target signals in a world where one person's nontarget signal might look like another person's target signal [9].

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The P300 speller is one of them, as individual subjects' P300 responses differ significantly from each other [10]. Although the P300 response is elicited approximately 300ms after a target stimulus, the actual latency, as well as the amplitude and shape, differ greatly among individuals. But with a sufficiently large inter-stimulus interval (ISI), such difference is not of great importance. However, applications such as the P300 speller have a small ISI with which to increase the information transfer rate, thus making it susceptible to individual differences [11].

In order to overcome this problem, traditional ERP-based systems have used machine learning approaches [12]. Although these systems perform satisfactorily, they often require calibration or a training session before use, because

supervised algorithms require labeled data for training [13]. During these sessions, sample data are extracted from the subject, and a corresponding classifier is constructed. Throughout this tedious process, the subject is often instructed to focus on a specific set of letters without any meaningful interaction. This calibration process is one factor among many that limits the practical use of the current BCI systems.

Various attempts have been made to reduce or eliminate the calibration stage [14]. In 2009, Lotte *et al.* proposed a regularized canonical correlation analysis combined with a regularized linear discriminant analysis in order to reduce the calibration time [15]. In 2011, Rivet *et al.* proposed an adaptive training session to achieve the same goal [16]. Although individual P300 signals may differ from each other, they also share features that vary less significantly. In order to capture such stationary features, information geometry, such as Riemannian geometry, has also been tried to help increase performance and to reduce calibration time [17], [18].

Many of these methods rely on data previously acquired from other subjects [12], [19]. Beginning with a generalized classifier, these methods update weights when subject-specific data comes in [21]. Other components, such as error-related potential, are sometimes used to determine the true label [22] in the concept of semi-supervised learning. These methods are useful because the classifier can provide a meaningful output when no previous subject-specific data is present. However, these classifiers still perform at relatively low levels in the beginning before they adapt to the new subject [23].

Conversely, there are other methods that do not require any kind of historical data. One obvious approach is to find a distinctive signal within the trial block. Since we know that there will be distinctive trials inside each block, classifiers can be used that are sensitive to such differences [24]. However, because of the low signal-to-noise ratio, task and data constraints are also used to classify the data, and supplementary components (e.g., error-related potential) can be combined in order to update classifier weights during usage of the BCI system [25]. Other studies have used expectation-maximization algorithms [26], [27], learning from label proportions [28] as applied in machine learning [29], or have used both model estimators simultaneously [30]. There also are methods that instead rely on paradigm design [31], and use prior knowledge given in the paradigm.

Combined, these approaches can sometimes produce better results. In recent studies, Kindermans *et al.* [32], [33] proposed a framework that adapted multiple techniques, including language models. Their work also incorporated transfer learning and unsupervised adaptation. In their simulation study, with a relatively large data sample (approximately 63,000 samples from 22 subjects), they demonstrated that the combination of these methods worked very well for calibration-less BCI. Other methods not aimed at eliminating calibration time completely seek to drastically reduce it [34]. In Qi *et al.*'s work, Riemannian distance was used in order to select similar ERP samples from an existing dataset, resulting

in a calibration time of 12 seconds [35]. Other works have tried to estimate whether a person will perform well with a limited amount of data [36], [37].

Although these methods perform relatively well compared to conventional supervised methods, they often initially perform relatively poorly due to insufficient data, and are still not fully comparable to traditional state-of-the-art methods in terms of accuracy, especially when insufficient subject-specific data are fed. Few studies have been tested in an online BCI situation.

An ideal calibration-less algorithm should take one of two paths: (1) determine a set of perfectly generalized features that do not differ among subjects; or (2) store various and infinite forms of P300 signals for comparison. Either approach is ideal, although neither is actually possible in the real world. There is no perfectly common feature, and one subject's target signal may look like someone else's non-target signal. Indeed, the perfect algorithm rests somewhere between the two. It must have a reasonably generalized model (generalization) and be able to capture various forms of signals (robustness).

Recently, convolutional neural networks (CNNs) [38] have been proven to be useful in distinguishing neurophysiological features [39]. Since the development of a CNN, various forms have been studied and optimized for EEG classification [40]. Because of the model's ability to capture local features, and thus being sufficiently robust to data shifts, recent architectures have been tested for inter-subject accuracy as well [41]. Because CNN can store "what it has seen" and derive a set of weights that generalize relatively common features, we hoped to determine whether CNN could identify the optimal point between the two paths above (generalization and robustness) in order to achieve calibration-less BCI.

In this context, the primary goal of this study is to see if a neural network, specifically CNN, could achieve zero-training in the P300 BCI system when fed a large dataset from different subjects. Thus, in this work, we fed a large data sample (99,000 samples) acquired from 55 subjects into a CNN in order to accomplish a calibration-less BCI, also referred to as zero-training. We focused on testing whether such an approach could work in a zero-training paradigm. With this aim, well-known and verified CNN architecture were used [41]. If it worked, our next questions would be: "How much data would be needed in order to achieve optimal performance?" and "What kind of data would be suitable to be used as a training set for such an approach?" We investigated these issues through a thorough analysis, and to the best of our knowledge, this is the first study that has attempted to use a CNN with such a large number of ERP samples from multiple subjects in order to achieve zero-training in a P300 BCI, as well as to investigate the impact that the amount of data has on the performance of neural network-based transfer learning in zero-training. We also tested it online using a P300 BCI speller system in order to see if our model worked in the real world.

The remainder of the manuscript is organized as follows. First, we investigate the CNN's ability to achieve zero-training with a large data sample by comparing it to the conventional approach using the subjects' data to train a classifier (e.g., stepwise linear discriminant analysis, or SWLDA). With this goal, our study begins with an offline experiment through which we show that a CNN can achieve reasonable performance only by using other subjects' data. Next, we conduct further investigation in order to determine how the zero-trained CNN works by investigating the amount of data needed to use such an approach, followed by investigating which data would be suitable for such an approach. The results from the online experiment are presented following the evaluation. Finally, we illustrate how a zero-trained network can serve to reach optimal performance in the form of semi-supervised learning [20], [42]. In this study, we use the term "live-training" to refer to a new classifier constructed with a label from both the zero-trained network and the subject's own input data.

II. MATERIALS AND METHODS

A. EXPERIMENTAL SETUP

The offline analysis was conducted on data acquired from 55 healthy subjects (male: 41, female: 14) [43]. The Institutional Review Board of Gwangju Institute of Science and Technology approved the experiment, and all experiments were performed following relevant guidelines and regulations. Before the actual experiment, we collected subjects' signed consent forms. Informed consent was obtained from all participants, or from their legal guardians.

Each subject went through the conventional 6-by-6 P300 speller paradigm, in which they were instructed to focus on a specific letter while one of six columns or rows flashed in a random sequence [44]. When the letter's row or column intensifies to white, the P300 signal is elicited from the parietal lobe. This signal was then measured with an electroencephalography (EEG) device, "BIOSEMI ACTIVE2," through BCI2000 [45] from 32 channels using the international 10-20 system at a 512Hz sampling rate.

Each subject underwent two training sessions with five letters ("BRAIN" and "POWER"), and four testing sessions with seven letters ("SUBJECT," "NEURONS," "IMAGINE," and "QUALITY") using BCI2000's default P300 speller and its classifier (Stepwise Linear Discriminant Analysis: SWLDA) [46]. SWLDA adds a feature extraction process before it performs the linear discriminant analysis (LDA), and has been used widely in P300 paradigms [9]. It has also outperformed other methods in various studies [9], [47], and hence was chosen as the method to be compared with. We implemented SWLDA with a "stepwisefit" function in MATLAB. Setting the maximum p -value for a term to be added to 0.1, we chose the best 60 features in terms of a small p -value.

As EEG signals contain noise components, each letter requires several trials for detection, and noise components are

reduced when these trials are averaged [9]. In our case, each letter was presented in 15 trials. The duration of intensification or flicker was 62.5ms, and the time between each flicker, often referred to as the ISI, was 125.5ms. Approximately 33 ($15 \times 12 \times 188\text{ms}$) seconds were required in order to spell one letter. Overall, we collected 99,000 training trials and 277,200 test trials from the 55 subjects.

B. PREPROCESSING

The data acquired from the experiment were processed as follows:

1. Each channel was re-referenced using the common average reference (CAR).
2. Data were filtered to the frequency band of interest, in this case 0.5Hz $\sim$ 10Hz, using the FieldTrip toolbox [48] with the 4th order Butterworth filter.
3. Data were segmented to 200ms $\sim$ 600ms epochs from each stimulus onset. A baseline correction was performed using the average value of $-200\text{ms} \sim 200\text{ms}$. Normally, baseline correction is performed using values before the stimulus onset. In this manuscript, we excluded signals before 200ms and used them for baseline correction, because P300 components are found typically at approximately 200ms $\sim$ 600ms [49]. As the ISI was set to 188ms, there is a non-target sample that looks like a target sample shifted by 188ms. The baseline was extended in order to reduce its effect on the baseline and to exclude VEP (visual evoked potential) signals. This adjustment also yielded the best performance in our research.
4. Data were down-sampled to a 128-sampling rate, which reduced the number of data from 512 time samples per second to 128 time samples per second for each trial. Not only did it produce the best results, but we also tried to keep the parameters the same as its original author. For the linear classifiers used for comparison, the downsampling rate was 20 Hz. We tested various preprocessing parameters and eventually decided to use the different sampling rates at which each algorithm performed best.

C. DEEP LEARNING MODEL

We used a CNN for classification for the zero-trained network. CNNs are known for their robustness to noise and shifts in data, as well as their ability to extract local features [50]. Although it originally gained its reputation in image recognition, it has recently proven to be effective in BCI research as well [51]. As each individual's P300 response differs largely in latency and amplitude [10], we chose a CNN, as we expected it to be more robust to these components.

The focus of this research was not on which architecture the CNN should have. Previous work has shown that a CNN can recognize ERP signals as well as other algorithms [39], and a recent study also provided us with architectural guidelines by which to construct an efficient CNN network for EEG analysis [40]. Instead, our focus was on the ways that

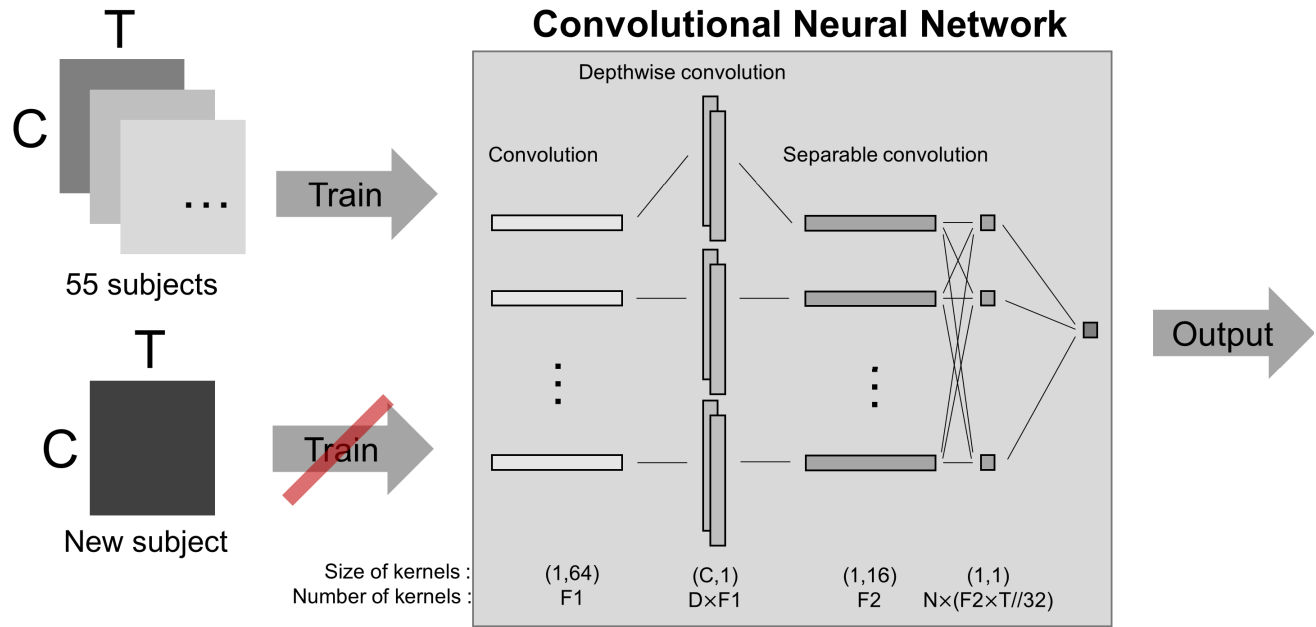


FIGURE 1. CNN architecture overall. C is the number of channels; T is the number of time samples; F_1 and D were 64 and 4, respectively, in this work; $F_2 = F_1 \times D$ as suggested in the original EEGNET architecture.

these networks can generalize different P300 signals from different subjects, specifically when using a large dataset.

We used EEGNET in this paradigm, which is a compact CNN that has been referenced various times since its first appearance [41]. Its distinct feature is its ability to be applied to various paradigms and to visualize interpretable features, and it also requires less data for training compared to other architectures [39], [40]. Its performance was validated with several state-of-the-art methods in the original paper as well, so we considered within-trained EEGNET as a state-of-the-art classifier together with SWLDA.

There were some parameters within EEGNET that we had to choose before usage concerning the number of kernels for each layer. In the EEGNET's original paper, the F_1 parameter controls the number of feature maps generated from the time domain, whereas D controls the number of spatial filters for each map. One would need to adjust these parameters according to the dataset provided. The recommended F_1 and D are both small values, less than eight in the original paper. However, since we thought that the network would need a huge number of parameters in order to capture various forms of ERP signals, we increased these parameters and observed how performance subsequently changed. Indeed, accuracy increased as the number of parameters increased. It did not matter whether F_1 or D increased, as long as $F_1 \times D$ increased. This increase became minimal after a certain amount, and the only additional costs were in more computation time. Therefore, we set F_1 to 64 and D to 4, as it yielded optimal performance. In the case of within-trained CNN, we set F_1 to 4 and D to 2, as recommended in the original paper.

The architecture of our CNN and zero-training approach is shown in Fig. 1. Although EEGNET has its own distinctive

characteristics, note that other architectures in the CNN family should provide similar results with our approach. Below are the brief details of CNN architecture that we used in this study.

- Layer block #1: 2-dimensional convolution layer followed by batch normalization layer [52]. F_1 number of feature maps will be generated.
- Layer block #2: Depthwise convolution layer [53]. It was primarily used in computer vision in order to reduce the number of training parameters. Setting the depth to D will create D number of spatial filters for each feature map. This layer is followed by another batch normalization and ELU (exponential linear unit) layer [54]. 2-dimensional average pooling is done in order to reduce the number parameters. The dropout technique is done to regularize the model [55].
- Layer block #3: The separable convolution layer is basically pointwise convolution [56], and this is done after depthwise convolution. The rest is the same as Layer 2, with a softmax layer at the end for classification.

We randomly selected 5% out of training data and set them as the validation set. For the zero-training method, if a subject's data has been used in the training set, the subject is excluded from the test set. We trained the network with the Adam [57] optimizer with the same parameters in its original paper, and set the validation patience to 5. Other than F_1 and D , we used the parameters recommended by the original paper [41]. We set the kernel length to 51, which is approximately 400ms in a 128 Hz sampling rate, and the dropout rate to 0.25. We also set a maximum norm for the incoming weights in the dense layer to 0.25, as recommended.

The models were trained with two NVIDIA GeForce GTX 1080 Ti GPUs using Keras API [58] and tensorflow [59]. When training on all training samples from 55 subjects (99,000 samples), it takes less than 5 minutes to train the network. When using the test trials as well (376,200 samples), it takes approximately 15 minutes to train the network. The size of the training and the testing data combined was 68 Giga bytes.

D. PERFORMANCE EVALUATION

Each subject performed two training sessions with five letters and four testing sessions with seven letters, as mentioned in the first section. Because there were 55 subjects, we had 99,000 training samples from which to train. For the zero-training methods, we used the leave-one-out method, in which we excluded the subject's data and trained CNN with the remainder of the training samples. For the within-training methods, only the subject's data ($n = 1,800$) were used for training. The trained classifier was tested equally for both approaches in four test sessions, and performance was measured as letter detection accuracy - the number of letters that the classifier identified correctly out of all 28 letters in the test session. For the letter spelled to be correct, the classifier had to obtain both the correct row and the correct column at the same time. The likelihood of obtaining the correct letter by chance is less than 3%.

Accuracy was also measured when fewer iterations were used instead of the 15 iterations. Accuracy increased as iterations increased per letter, but it took longer to spell one letter as well. If we used all 15 iterations per letter, it would take approximately 33 seconds, whereas using one iteration would take only 2.2 seconds. Offline analysis was conducted using only N number of iterations for letter accuracy detection; we selected the first N number of samples out of 15 samples in each letter (column/row). Using five numbers of iterations would take 11.2 seconds, and 10 numbers of iterations would take 22.5 seconds per letter. All four methods are compared: SWLDA_WT (within-trained SWLDA), SWLDA_ZT (zero-trained SWLDA), CNN_WT (within-trained CNN), and CNN_ZT (zero-trained CNN).

E. STATISTICAL TEST

The mean and the standard deviation were calculated from each distribution and are reported in the text. A paired t -test was conducted, and significance was evaluated as $p < 0.05$, < 0.01 , and < 0.001 throughout this study.

F. ANALYSIS OF SUFFICIENT DATA

In order for the zero-trained network to show reasonable accuracy, the CNN must examine various samples from various subjects. The question remains as to how many subjects were needed to obtain data from. We determined the number of subjects needed in order for the network to reach its full accuracy through an offline analysis, which was accomplished by omitting a subject's data as a test set and using subjects' data selected randomly as the training set. The

number of subjects selected randomly was increased from 1 to 54, as we had a total of 55 participants. For computational purposes, we tried only 30 combinations selected randomly. It is nearly impossible to compute every possible trial, because it would take ${}_{55}C_n$ trials to compute for n number of subjects. The combination selected randomly did not contain the test subject's data, and this computation was performed with all 55 subjects in order to determine their mean accuracy.

The question remains: is the increase in accuracy attributable to large data or diversity? In order to address this issue, two additional tests were performed. The first limits the number of training data while also increasing the number of subjects from whom the training data are selected. In this way, we were able to measure the diversity in the samples' effect. The next method is to fix the number of subjects used for training and to measure accuracy while also increasing the dataset's size. Ideally, this should be examined first using only one subject's data. However, the number of data from one subject is limited, and such a method is not expected to yield reasonable accuracy regardless of the dataset's size. Instead, we limited diversity to a certain level, and increased the data size accordingly. Data were selected randomly for each computation for both methods.

G. ANALYSIS OF SUITABLE DATASET

In order to see what kinds of data produce better results in transfer learning, or what kinds actually hinder performance, we performed three additional analyses.

We first investigated which factors contribute to high performance in a within-training paradigm by examining letter accuracy along with P300 latency and amplitude. These components were measured after averaging all of the target samples from the training set.

Secondly, we calculated the average of the P300 amplitude and latency for each subject. Then we performed one-on-one transfer learning on all 55 subjects. Excluding the within-trained cases, this resulted in 55×54 cases. With every combination, the difference in latency of the two subjects and the corresponding transfer learning accuracy were measured. As the latency difference was continuous, we divided these values into 10ms discrete sized bins and found the mean accuracy for each bins. The same investigation was applied with the difference in P300 amplitude, the bin size being 0.1 microvolts.

Lastly, we picked the best subject for transfer learning for each testing subject. This is the subject that produced the highest accuracy when trained with for the testing subject. We then added another subject's data in order to see whether the accuracy would increase or decrease. In other words, we picked two subjects' data to be used for transfer learning; one that has produced best results for the subject, and one that we wish to compare with the testing subject's data. This was done to see whether adding more data always increases the network's performance, as well as to examine what kinds of data would contribute to such differences. All of the tests

were done on CNN-ZT architecture. Excluding the testing subject's data and the initial subject's data that is used for transfer learning, this results in 55 (target subjects) $\times$ 53 (extra subjects) cases.

H. ANALYSIS OF MODEL SIMILARITY

The ability to explain features was an area of common interest in previous research on using a CNN with EEG data. Specifically, researchers were interested in whether such networks could detect neurophysiological and interpretable components of EEG data, as this could be an important issue for neuroscientists. Despite the old belief that neural networks often are "black boxes" concerning interpretation, previous studies have demonstrated various methods with which to overcome such issues [60], [61].

In this research, our focus was on how a CNN could effectively generalize P300 data when trained with a massive dataset, compared to traditional methods when only the subject's individual data were used. In order to do so, we employed two methods. First, the output scores that each model produced for the within- and zero-trained methods were compared. A Pearson's correlation analysis was conducted on the series data (output score) from two models, with the goal of determining whether the two models (within- and zero-trained) produced similar outputs. The second method we used is similar to that proposed in [28]. It was primarily used to examine how correlation increases with each deep learning layers. However, rather than correlating the input-feature and unit-output features, we examined the relevance by looking only at the correlation between the input-feature and output score calculated with 5,040 test samples for every subject, which allowed us to examine the network's behavior as a whole. Because each feature produces its corresponding feature-to-output relevance or correlation map, we further calculated the similarity of the relevance between the within- and zero-trained models by correlating the two relevance series. By performing correlations on the correlation map produced by the two different approaches, we could obtain the final model similarity value for each subject. A high correlation value would indicate that the zero-trained network behaved in a similar fashion as the within-trained network, thus meaning better generalization. The same was done for SWLDA for comparison.

I. ONLINE STUDY

In order to determine the true performance of a BCI system, an online test is needed in which feedback is presented to the user. To validate the applicability of the CNN_ZT constructed, an online test was conducted with different subject groups. This experiment was approved by the Public Institutional Bioethics Committee designated by the MOHW. The same paradigm was used as in the offline study in order to match the conditions for fair comparison. Twelve new subjects participated in this online study and played the P300 speller to confirm the ability to generalize the zero-trained CNN. The CNN_ZT model constructed previously

was tested with each new subject with the same paradigm. In order to obtain an accurate comparison, the subject was instructed to spell 14 letters in the zero-training mode (using CNN_ZT) and another 14 letters using the conventional P300 speller; SWLDA (trained with the subject's individual data) was used as the conventional method, with a training session of 10 letters just before the conventional online spelling. In order to eliminate the influence of accumulated fatigue, we switched the order of these two approaches on half of the subjects. On either case, subjects underwent a training session just before performing the conventional testing session. The results appear in Table 2.

Twelve healthy subjects (eight males and four females) participated in the experiment in order to confirm the feasibility of our zero-training approach. Each subject underwent four testing sessions of seven letters ("SUBJECT," "NEURONS," "IMAGINE," and "QUALITY"). Of these four sets, two used our zero-training approach (CNN_ZT), and the other two used the conventional approach (SWLDA_WT). Because the conventional approach requires a calibration session, we conducted two training sessions of five letters ("BRAIN" and "POWER") just before the SWLDA_WT session. Six subjects attempted zero-training first, and the other subjects attempted the conventional method first in order to eliminate the influence of fatigue and any sequential effect. Then, letter detection accuracy was calculated and compared with each method.

J. LIVE TRAINING APPROACH

This section briefly describes an example of the way our zero-trained network can be manipulated to further enhance its accuracy. The live-trained network itself is a simple, semi-supervised adaptive learning algorithm that requires no training phase. The concept is similar to the online adaptive classifier used by Kindermans *et al.* [42] that updates itself whenever new data comes in. Whereas supervised classifiers typically obtain their labels from outside of the system, the live-trained network obtains the data label from the zero-trained network. The label acquired here would not be identical to the true label, as the zero-trained network has some error. The SWLDA was used as the classifier for the live-trained network in this study.

Expectedly, the live-trained network's initial level of accuracy was extremely low, as it did not have enough samples to train itself. The network is updated every time a new letter is entered, and in this P300 speller paradigm, it takes 15 trials to spell one single letter. Therefore, after spelling one letter, it would have 180 (15×12) more samples on which to train from, of which 30 (15×2) samples would be targets, and the remainder non-targets.

From the live-trained network perspective, this is the same as being given the subject's data and label. However, some label noise is included, and some labels from another model may be wrong. Thus, it is always best to use the subject's individual data for training. We approached this problem as a label noise problem in machine learning [62] and used

TABLE 1. Comparison of all 55 subjects' performance.

Subject #	CNN_WT	CNN_ZT	SWLDA_WT	SWLDA_ZT	Subject #	CNN_WT	CNN_ZT	SWLDA_WT	SWLDA_ZT
1	85.71	89.29	92.86	71.43	30	100.00	75.00	100.00	67.86
2	82.14	42.86	96.43	39.29	31	67.86	85.71	82.14	82.14
3	96.43	100.00	100.00	78.57	32	100.00	96.43	100.00	75.00
4	100.00	92.86	100.00	67.86	33	100.00	100.00	100.00	100.00
5	96.43	92.86	100.00	60.71	34	96.43	92.86	92.86	89.29
6	100.00	100.00	100.00	96.43	35	100.00	100.00	100.00	96.43
7	100.00	100.00	100.00	85.71	36	96.43	89.29	96.43	75.00
8	67.86	75.00	96.43	39.29	37	100.00	75.00	100.00	17.86
9	82.14	92.86	85.71	67.86	38	100.00	100.00	100.00	100.00
10	100.00	100.00	100.00	64.29	39	85.71	96.43	89.29	82.14
11	100.00	92.86	92.86	96.43	40	96.43	89.29	100.00	64.29
12	96.43	85.71	96.43	39.29	41	92.86	96.43	96.43	85.71
13	100.00	89.29	100.00	60.71	42	92.86	96.43	92.86	82.14
14	67.86	50.00	60.71	17.86	43	96.43	71.43	100.00	25.00
15	100.00	100.00	100.00	71.43	44	96.43	82.14	85.71	46.43
16	100.00	100.00	100.00	82.14	45	64.29	57.14	50.00	35.71
17	89.29	89.29	89.29	67.86	46	100.00	92.86	89.29	89.29
18	92.86	85.71	96.43	53.57	47	100.00	100.00	100.00	100.00
19	100.00	100.00	100.00	75.00	48	100.00	100.00	92.86	82.14
20	100.00	100.00	100.00	96.43	49	100.00	96.43	100.00	89.29
21	96.43	89.29	100.00	25.00	50	100.00	100.00	96.43	96.43
22	100.00	100.00	100.00	100.00	51	92.86	96.43	92.86	64.29
23	100.00	100.00	100.00	82.14	52	89.29	67.86	100.00	60.71
24	100.00	100.00	100.00	82.14	53	100.00	96.43	100.00	96.43
25	46.43	57.14	42.86	32.14	54	100.00	82.14	100.00	64.29
26	92.86	75.00	92.86	67.86	55	100.00	100.00	100.00	89.29
27	100.00	85.71	89.29	21.43	AVG	93.64	89.22	94.29	69.81
28	96.43	89.29	96.43	85.71	STD	11.16	13.65	11.51	23.57
29	92.86	96.43	100.00	53.57	MIN	46.43	42.86	42.86	17.86

the SWLDA classifier, as we expected it to be relatively robust to label noise. We also expected the classifier to be robust to noise only when there were enough training samples regarding the number of feature dimensions.

As illustrated in Fig. 6A, the final system proposed outputs at the same output as the zero-trained network until enough data are accumulated, as the live-trained network would lack sufficient samples from which to train. When the live-trained network acquires enough samples and data labels from the zero-trained network, the system uses the output from the live-trained network as its output.

This system requires no training phase. Instead, the predicted label from CNN_ZT is used to train SWLDA parameters, as mentioned. Further description is below:

Step 1. The initial prediction of n^{th} letter of the live-trained network is determined by CNN_ZT. Based on that information, create labels $y_{n,j} \in \{0, 1\}$ to the given input vectors $x_{n,j}$. j is the epoch number in the k^{th} letter.

From k^{th} letter, repeat Step 2 and Step 3:

Step 2. With the acquired ERP samples and their predicted labels until $n - 1^{\text{th}}$ letter, create training set s :

$s_{i,j} = (x_{i,j}, y_{i,j})$ with $1 \leq i \leq n - 1$ and $1 \leq j \leq M$, select the significant features, and train SWLDA parameters (w and b) for a decision hyper-plane defined by $w \cdot x + b = 0$.

Step 3. The letter is determined based on the predicted row and column of SWLDA.

$$row_n = \underset{row}{\operatorname{argmax}} \sum_{r=1}^{\text{repetition}} w \cdot x_{n,r,row}$$

$$col_n = \underset{col}{\operatorname{argmax}} \sum_{r=1}^{\text{repetition}} w \cdot x_{n,r,col}$$

where w corresponds to trained parameters of SWLDA. In this study, k and M were set to 4 and 180.

III. RESULTS

A. PERFORMANCE EVALUATION (OFFLINE STUDY)

Performance was measured as letter detection accuracy (i.e., the number of letters that each classifier obtained correctly out of all 28 letters). CNN and SWLDA were trained using both the zero-trained and a conventional approach (trained within a subject) for comparison, as described in the Method section. 'AVG' in the bottom of Table 1 shows the mean accuracy across the four methods: CNN_WT (93.63%), CNN_ZT (89.22%), SWLDA_WT (94.28%), and SWLDA_ZT (69.80%), respectively. Table 1 presents the individual subjects' classification accuracy across the four methods.

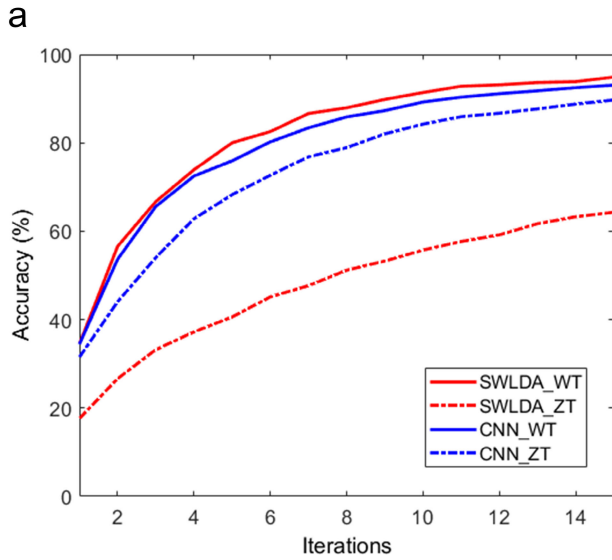


FIGURE 2. Average letter detection accuracy when the number of iterations is limited.

Average, standard deviation, and min value are shown at the right bottom corner. When the conventional approach was used, CNN_WT showed statistically similar performance to SWLDA_WT ($p > 0.05$), which is used widely in P300 BCI systems. Interestingly, the zero-trained network (CNN_ZT) exhibited comparable performance to the conventional state-of-the-art methods in the case of CNN_WT, whereas SWLDA_ZT did not. The low performance of SWLDA_ZT is in accordance with previous studies which also used SWLDA_ZT as a baseline to compare with [63]. However, the performance of CNN_ZT remained relatively low compared to CNN_WT or SWLDA_WT, and showed a difference of -4.41% (with CNN_WT) and -5.06% (with SWLDA_WT). A paired t -test (55 subjects) revealed that there is a significant statistical difference when comparing CNN_ZT with CNN_WT ($p < 0.05$) and SWLDA_WT ($p < 0.05$). Although the difference may look small, we believed that there was still room for further improvement, a topic that we return to later.

Fig. 2 shows what would happen if fewer iterations were used for classification, instead of 15. Accuracy is indicated in each case. If only one iteration was used, it would take only 2.2 seconds per letter. Likewise, five iterations would equal to 11.2 seconds, and ten iterations would equal to 22.5 seconds. When given a single sample, the three methods (CNN_WT, CNN_ZT, and SWLDA_ZT) perform relatively the same, and show no statistical difference with each other. In the cases of 5 and 10 iterations, there was a statistical difference between within-trained classifiers and CNN_ZT ($p < 0.05$), but overall difference remained below 10%.

B. ANALYSIS OF MODEL SIMILARITY

In order to determine whether CNN effectively generalizes a model that works for different subjects, we investigated

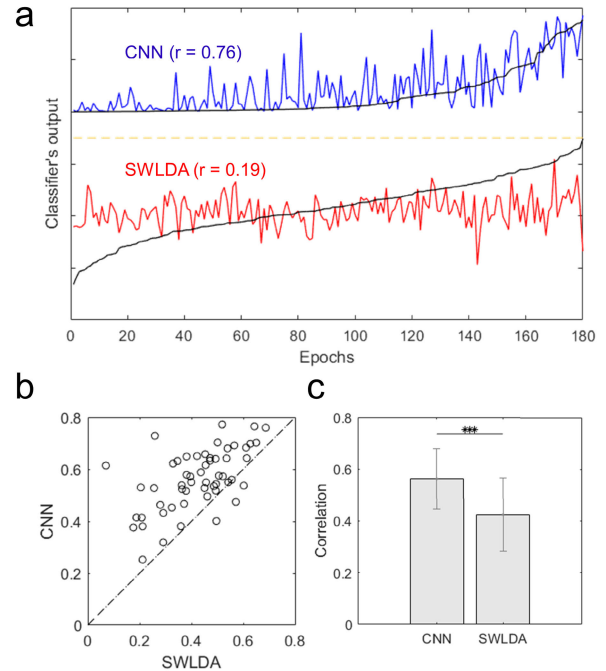


FIGURE 3. Output score comparison for different classifiers. (A) Classifiers' outputs over epochs from one representative subject (s37) appear in the upper (black line: CNN_WT, blue line: CNN_ZT) and lower sections (black line: SWLDA_WT, blue line: SWLDA_ZT). CNN_WT and CNN_ZT were correlated more highly ($r = 0.76$) than were SWLDA_WT and SWLDA_ZT ($r = 0.19$). (B) Correlation values from all 55 subjects are presented: the x-axis is for SWLDA, and the y-axis is for the CNN. (C) The mean and standard deviation of the correlation values are plotted. The significance level is marked with an asterisk (***) $p < 0.001$.

the classifier's output score. Fig. 3A shows the output scores of the zero-trained approach and the within-trained approach for SWLDA and CNN from one representative subject. Over the epochs, SWLDA_WT and SWLDA_ZT did not have a strong correlation ($r = 0.19$), whereas CNN_WT and CNN_ZT showed a relatively high correlation ($r = 0.76$), indicating that the classifiers' output patterns were similar. The correlation coefficients from all 55 subjects appear in Fig. 3B. Most points lay above $y = x$, indicating that the CNN generalized better than did the SWLDA. Summarizing these results, the similarity in output scores between the zero- and within-trained approaches was higher ($p < 0.001$) in the case of CNN (mean $r = 0.56$) than that of SWLDA (mean $r = 0.42$), as shown in Fig. 3C.

We also investigated the relationship between the feature and the output score. Fig. 4A shows the feature-output score relevance in one representative subject obtained from the correlation analysis. We did this by correlating output scores from the model with input feature values on all of the samples (see Materials and Method for additional detail). The bottom section of Fig. 4A illustrates the corresponding topographic image using the mean value over the time samples. The patterns were very similar across the three methods (SWLDA_WT, CNN_WT, and CNN_ZT), whereas SWLDA_ZT differed from the others. We also quantified

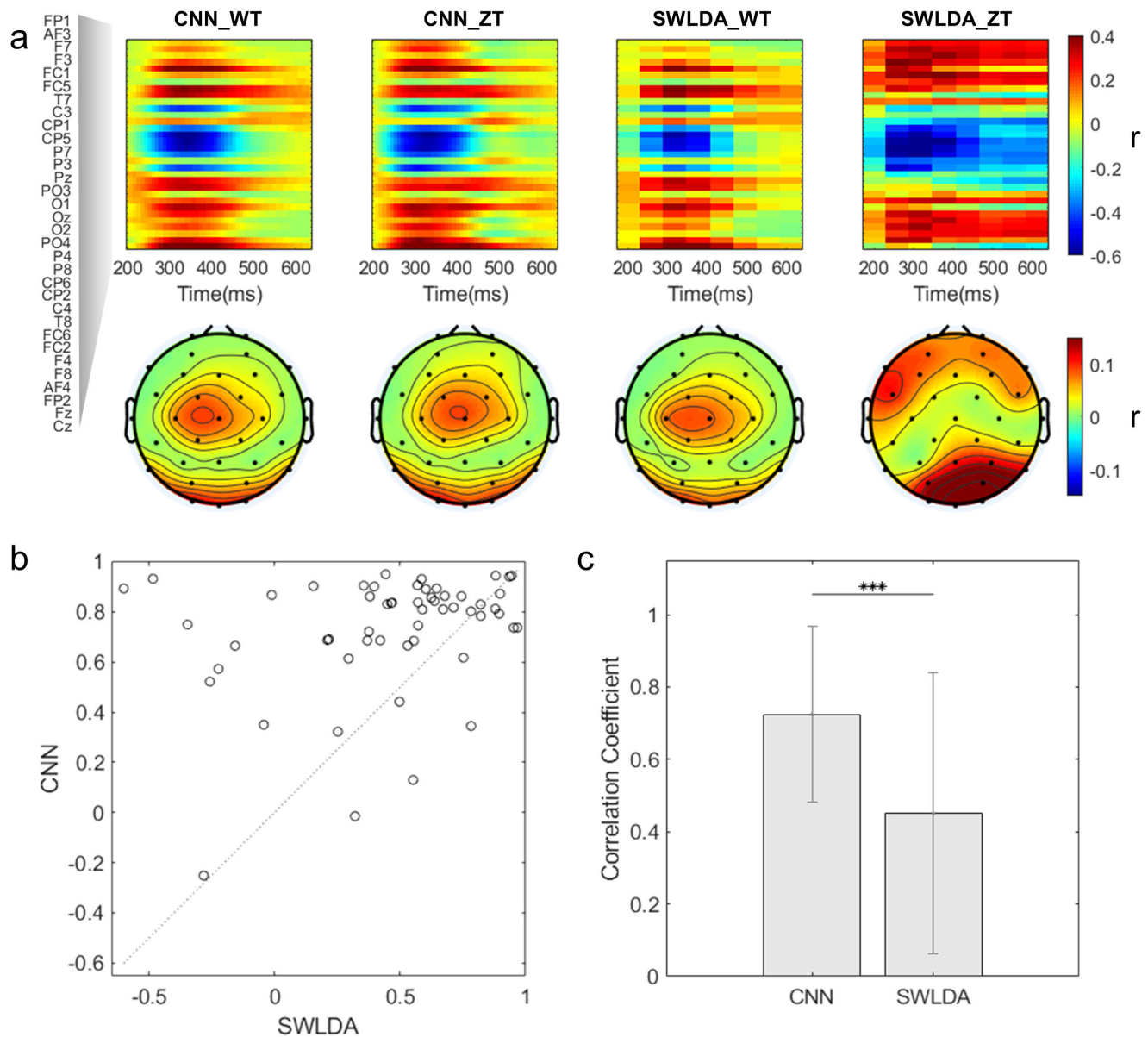


FIGURE 4. Feature-to-output relation calculated with Pearson correlation analysis. (A) The correlation map from one representative subject is shown for every method. The correlation coefficient (r) is inserted to show the similarity between the two images (CNN_WT vs. CNN_ZT and SWLDA_WT vs. SWLDA_ZT). The topographic image of the mean value over time samples is shown below as well. (B) The similarity values between the within-trained and zero-trained models are shown across all 55 subjects; the x-axis is for SWLDA, and the y-axis is for the CNN. (C) The mean and standard deviation of the similarity are presented with the significance value (***: $p < 0.001$).

the similarity value by correlating the two images shown in Fig. 4A. The representative subject showed high similarity ($r = 0.94$) in the case of CNN_WT and CNN_ZT, but only $r = 0.44$ in the case of SWLDA_WT and SWLDA_ZT. In order to summarize the results for all 55 subjects, their similarity values were collected as well (Fig. 4B). Indeed, we see a trend in which higher similarity exists when comparing CNN_WT and CNN_ZT, and relatively low similarity exists for SWLDA_WT and SWLDA_ZT. Statistical tests revealed that the mean similarity was significantly higher in the case of CNN than it was in SWLDA ($p < 0.001$), as shown in Fig. 4C. These results indicate that the CNN better

generalized an internal model that transforms the input feature into a reliable output score, even when different subjects drove the data training. The similarity between CNN_WT and CNN_ZT indicated that the network could react in the same way regardless of the presence of the subject's data, which is important in terms of achieving calibration-less BCI.

C. ANALYSIS OF SUFFICIENT DATA

Accuracy increased when we also increased the size of the training dataset by adding a subject's training data, which is equal to 1800 samples. The number of the subjects used for training, along with the corresponding accuracy, appear in

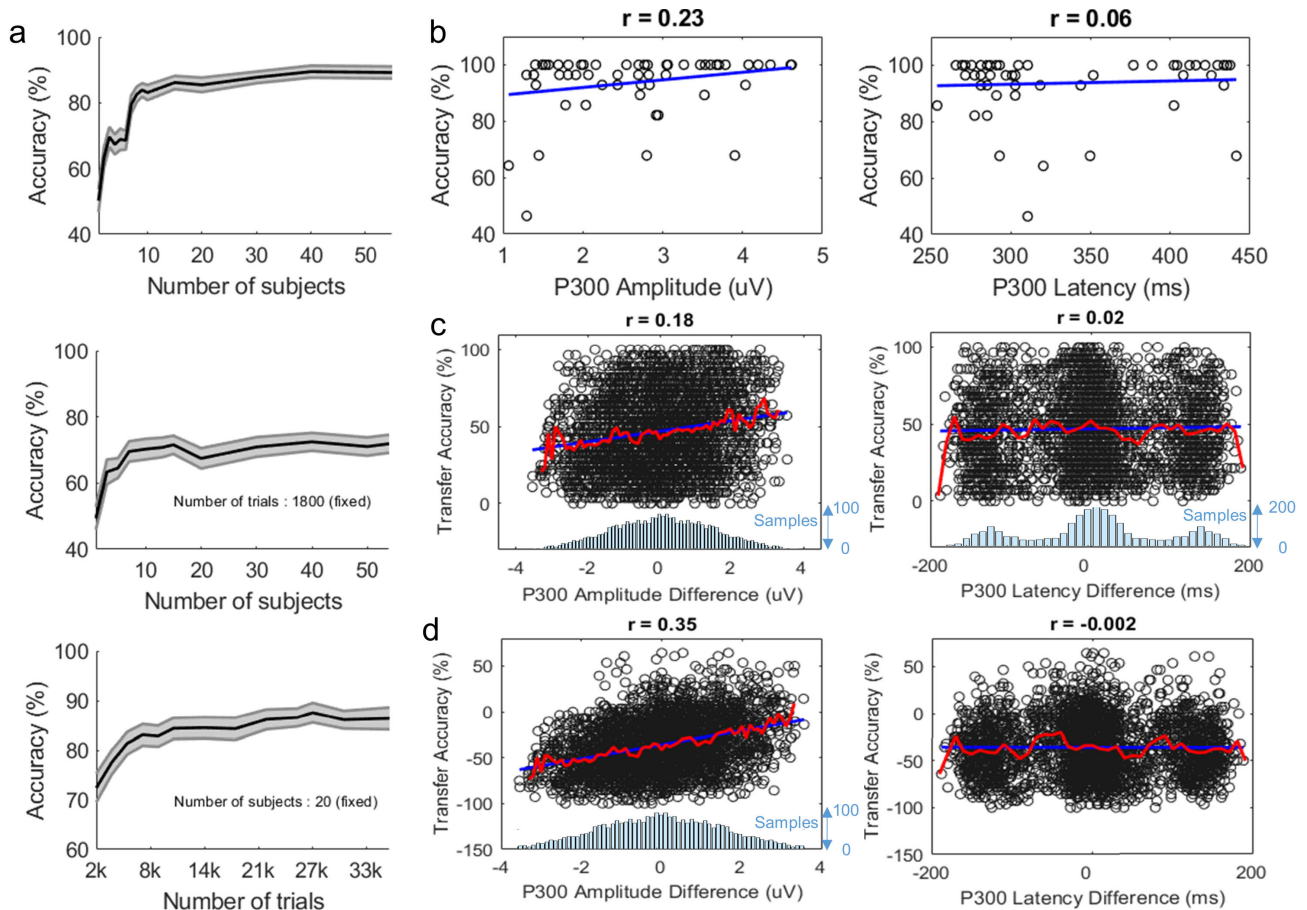


FIGURE 5. Performance changes. (A) The change in letter accuracy as the number of subjects used for training changes is on the left. Letter accuracy was measured as other subjects' data were added to the training set. As there are 1,800 training samples for each subject, adding one subject's data increased the dataset size by 1,800. The figures on the right show accuracy when the number of training trials (top right figure) or diversity (number of subjects) (bottom right figure) is fixed. (B) Letter accuracy (Within-trained) is measured along with each subject's average P300 amplitude and P300 latency. (C) One-on-one transfer learning accuracy is measured along with the difference in P300 amplitude and P300 latency. (D) The changes in accuracy when adding a subject's data is measured along with the difference in P300 amplitude and P300 latency.

the top figure of Fig. 5A. The performance increased as the number of training samples increased, and only a marginal change was observed after approximately 18,000 samples (10 subjects' data).

In the simulation above, the number of training trials, as well as the number of subjects, was increased, which raises another question: Is it the number of samples or the number of different subjects (diversity) that contribute to constructing a generalized CNN model? Although the top figure of Fig. 5A shows that the number of trials and diversity contributed to an increase in accuracy, it does not clearly indicate which factor is more important. Therefore, in order to distinguish between the two (amount of data and diversity), further simulations were conducted. First, we limited the size of the training data to 1,800 and increased the number of subjects that were used (the middle figure of Fig. 5A). Second, we fixed the number of subjects used to 20 and increased the training data (the bottom figure of Fig. 5A). As the figures show, performance increased in both cases, but the optimal performance shown in the top figure of Fig. 5A was not achievable. Thus, it appears

that both factors contribute to an increase in accuracy, meaning that not only do we need a large amount of data, but we also need a sufficiently large number of subjects.

D. ANALYSIS OF SUITABLE DATASET

As seen in Fig. 5A, adding more data did not always increase the performance of the network. There were some points when accuracy slightly decreased, while the size of the data increased. In order to investigate whether this is due to randomly selecting low-performing subjects or certain subjects hindering the performance of the network, we first investigated which factors (P300 amplitude or P300 latency) contribute to higher performance in within-training paradigm. This is shown in Fig. 5B. The regression line is shown as a blue line, and the Pearson correlation value is shown at the top. As we can see from the figure, the mean amplitude is related to accuracy ($r = 0.23$). This is quite expected, since a lower mean amplitude would indicate that the P300 signal is not elicited as frequently as it should be, or is small so that the classifier cannot easily distinguish it.

TABLE 2. Result of Online P300 Spelling Experiment (Bold Represents Zero-Training).

Subject #	Session 1		Session 2		Num. Correct letters (accuracy %)	
	SUBJECT	NEURONS	IMAGINE	QUALITY	Within-trained (SWLDA_WT)	Zero-trained (CNN_ZT)
Sbj1	SUBJECT	NEURONS	IMAGINE	QUALITY	14 (100%)	13 (93%)
Sbj2	SUZLEIT	NEWQNS	IMAGIN	QUALITY	9 (64%)	12 (86%)
Sbj3	SWBVECT	NEB4CWG	IMAFSNQ	9UMLHY	8 (57%)	6 (42%)
Sbj4	SCBJEET	TEURONF	IMAGINE	QUALITY	14 (100%)	11 (79%)
Sbj5	YUBJKCZ	TEUXUNN	IMGNITE	QUAXITY	7 (50%)	10 (71%)
Sbj6	SIBJEIT	NEURONG	IMAGINK	WUALITY	12 (86%)	11 (79%)
Sbj7	SUBJQCT	NE1RONS	IMGINE	QUALITY	13 (93%)	12 (86%)
Sbj8	SUBJECT	NEURONS	IMAMINE	QUALITY	13 (93%)	14 (100%)
Sbj9	SUBJECT	NEURONS	IMAGINE	QUALITY	14 (100%)	14 (100%)
Sbj10	SWBJEF6	NEURONS	IMSAIJK	QUALCTY	7 (50%)	13 (93%)
Sbj11	SUBJECT	NEURONS	OMAGINE	QUALITY	14 (100%)	13 (93%)
Sbj12	SUBJECT	NEURONS	IMAGINE	QUALITY	14 (100%)	14 (100%)
Total (Mean $\pm$ STD)					139 (82% $\pm$ 20%)	143 (85% $\pm$ 15%)

We then examined whether the same result is achieved for transfer learning. We performed one-on-one transfer learning on all 55 subjects, and the difference in latency/amplitude and the corresponding transfer learning accuracy was measured. Excluding within-trained cases, all 55×54 cases are plotted as dots in Fig. 5C. As the latency/amplitude difference was continuous, we divided these values into 10ms/0.1uV discrete sized bins, and found the mean accuracy for each bin. The mean accuracy (red line) and number of samples belonging to each bin (blue bar graph) are shown as well. The difference in P300 latency had almost no correlation ($r = 0.02$) compared to P300 amplitude ($r = 0.18$). The negative value of the x axis in the first figure indicates that the training subjects' average P300 amplitude is smaller than that of the testing subjects. We can see that if the training subject's average P300 amplitude is smaller than the testing subject's P300 amplitude, transfer learning does not tend to work as well.

We also examined whether the same principle holds when adding a subject's data to an existing dataset. The existing dataset only consists of the best-performing subject for each testing subjects. Their average was 82.85%, which is lower than what is shown in Table. 2 (89.22%). When adding a new subject, accuracy tended to decrease, as they were less suitable than the existing data as shown in Fig. 5D. This decrease in performance was extreme for subjects with a relatively lower average P300 amplitude. While subjects with higher P300 amplitude could increase the transfer learning accuracy up to 9.28% (the rightmost value of the red line), subjects with lower P300 amplitude could hinder the performance up to -78.57% (the leftmost value).

E. ONLINE STUDY

We performed an online test on 12 subjects in order to validate our approach. In Table 2, spelled letters are presented in the 2nd to 5th columns, and the number of correct letters and corresponding letter detection accuracy (%) is shown in the 6th and 7th column for the within- and zero-trained (written in bold and italic letter) classifiers, respectively. Wrong letters were marked with a strikethrough. Six of the twelve subjects showed better letter accuracy in the within-trained approach. Interestingly, Table 2 shows that 143 of 168 letters were correct using our zero-trained CNN, while 139 of 168 letters were correct using within-trained SWLDA. The twelve subjects' mean accuracy was 85% ($\pm 15\%$) for CNN_ZT and 82% ($\pm 20\%$) for SWLDA. Overall accuracy is relatively low compared to offline accuracy acquired from 55 subjects, but this is due to certain subjects performing extremely low. However, our zero-training approach produced results that are comparable to the traditional within-training approach, because it achieved even higher accuracy in the online test. There was no significant statistical difference between the two methods ($p > 0.05$).

F. LIVE TRAINING APPROACH (OFFLINE STUDY)

In the offline study, we observed that CNN_ZT was comparable to SWLDA_WT and CNN_WT, but its performance was still relatively lower than the other two ($p < 0.01$). Considering the nature of data dependency, this is unsurprising, as only the subject's unique data can perfectly represent his/her data characteristics. In this context, our zero-training approach can

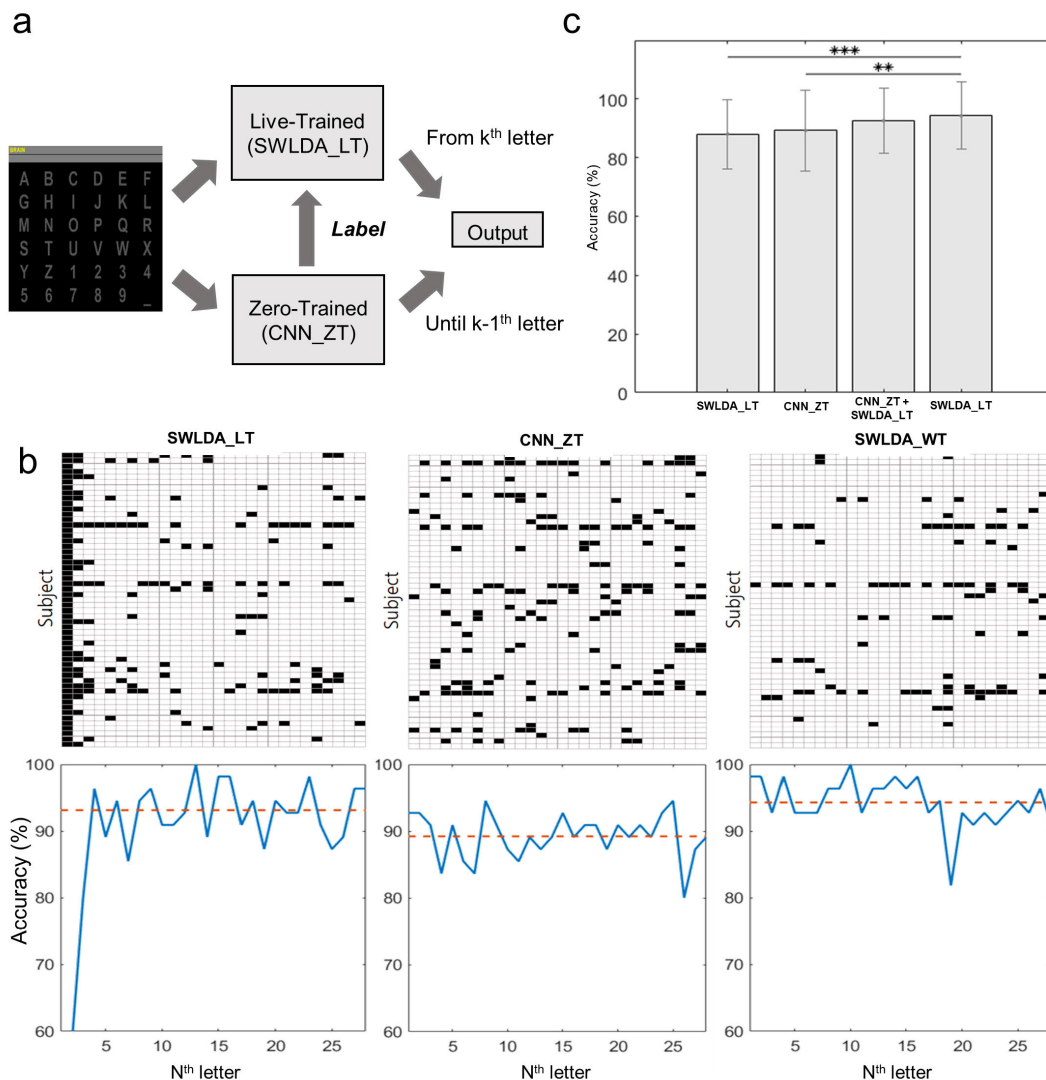


FIGURE 6. Live-training concept and simulation result. (A) The live-training system is illustrated. The label generated from the zero-trained model was re-used in order to train the conventional model. The zero-trained model was used until the $k-1^{\text{th}}$ trial, and was then switched to using the well-trained adaptive classifier. (B) Correctness (white) or incorrectness (black) is presented for each method (SWLDA_LT, CNN_ZT, and SWLDA_WT). Below each figure, the letter accuracy calculated for 55 subjects is presented over the N^{th} letter prediction. The vertical line in the figure for SWLDA_LT marks the first letter that shows no statistically significant difference between adaptive SWLDA and SWLDA_WT ($p > 0.05$). (C) This shows the letter accuracy for SWLDA_WT, CNN_ZT, and SWLDA_LT, together with the hybrid method. The method proposed (SWLDA_LT+CNN_ZT) achieves comparable performance to the within-trained classifier ($p > 0.05$), while the zero-trained approach still shows lower performance ($p < 0.01$). SWLDA-LT alone showed lower performance compared to SWLDA_WT ($p < 0.001$).

be manipulated further to increase the user's performance, and Fig. 6A illustrates this concept.

We conducted a simulation study of this semi-supervised learning, which we call live-training, as mentioned in the Methods section. SWLDA was employed again as the extra classifier trained while spelling. The simulation result appears in Fig. 6B and demonstrates that SWLDA_LT did not perform well in the initial prediction. However, as data were accumulated together with its estimated label, SWLDA_LT outperformed CNN_ZT ($p < 0.05$). As illustrated in Fig. 6A, the ideal solution is to use the output from CNN_ZT in the beginning, and then to switch to SWLDA_LT's output

beginning with the k^{th} letter, which is "live-training," the final approach proposed (SWLDA_LT: a hybrid of CNN_ZT and SWLDA_LT). In our case, we set the value k to 4. As Fig. 6C shows, SWLDA_LT+CNN_ZT (92.98%) and SWLDA_WT (94.28%) did not differ significantly ($p > 0.05$), indicating that SWLDA_LT combined with CNN_ZT reaches the ideal performance that one subject may achieve.

The number of data that would be enough for SWLDA_LT to be well-trained was also of significant interest. Although there are few fluctuations in the graph, a paired t -test (55 subjects) revealed that 540 trials (3 letters) was the

threshold at which the results did not differ significantly ($p > 0.05$). After three letters, a subject-specific classifier was constructed and was ready to be used in the conventional 6-by-6 P300 BCI speller paradigm, with 15 blinks per row/column.

IV. CONCLUSION AND DISCUSSION

In this paper, we have demonstrated a simple way to use other subjects' data to achieve performance similar to that of within-trained methods. We also proposed a hybrid system that further enhanced this approach by treating the entire problem as a label noise problem in machine learning [62]. We illustrated how a label generated from the zero-trained model could be reused to train the conventional model. Other noise-robust classifiers or systems could be utilized as well. Although training a neural network with a massive dataset could take a tremendous amount of time, this would not be a critical issue in the real world. Once the weights are trained, it works as fast as other classifiers in the field.

Previous works have achieved decent performance in zero-training [30], [32], and these methods could serve as a baseline for comparison. However, we set the performance of within-trained state-of-the-art classifiers as a baseline because this is the whole purpose of zero-training. One may think that regularized LDA (rLDA) may be a better baseline instead of SWLDA, as it can be robust to experimental settings and different preprocessing methods [64]. However, in our settings, SWLDA and rLDA showed no statistical difference in terms of letter detection accuracy: SWLDA: $94\% \pm 10\%$ and rLDA: $95\% \pm 10\%$ ($p > 0.05$). rLDA also showed no statistical difference when compared to the proposed hybrid method and within-trained CNN ($p > 0.05$). Hence, we set SWLDA_WT as a baseline method. Also, when replacing SWLDA with rLDA in the hybrid method and performing the same process mentioned in III-F, a letter detection accuracy of 94.35% was achieved. This also showed no significant statistical difference with within-trained rLDA ($p > 0.05$). We also set CNN_WT as a baseline method, because in the authors' original paper [41], it showed the same or superior performance when compared with the state-of-the-art methods mentioned in that work.

We also performed the same zero-training approach with rLDA as CNN_ZT, because SWLDA may not be a suitable candidate for cross-subject conditions. SWLDA is intended to reduce feature dimensions, and reducing feature dimensions when there are so many samples may hinder the generalization ability of the model. Regularization may be a more reasonable option, so we tried the same process with rLDA as well. rLDA_ZT (zero-trained rLDA) scored an average letter accuracy of 83.70% across 55 subjects. This is higher than SWLDA_ZT ($p < 0.05$), but still significantly lower than CNN_ZT ($p < 0.05$).

The dataset obtained and used here [65] contains a larger amount of data compared to previous works [33], [41]. It also has more diversity, as it came from more subjects. Most importantly, the focus of our research addressed how a neural

network (CNN) can effectively generalize signals from different subjects. Considering the complexity of a CNN, even larger amounts of data may boost the performance of the network. In Fig. 5A, there was little improvement even when a large amount of data was already fed into the network. What would happen if even larger amount of data was fed into the network was still of interest. We attempted this by including the test trials from all of the subjects. When training a CNN with all of the training data and testing data from other subjects, which is approximately 370,000 samples, CNN_ZT scored $92\% \pm 10\%$, and showed no statistical difference with CNN_WT ($p > 0.05$). However, it still showed some statistical difference when compared with SWLDA_WT ($0.01 < p < 0.05$). If even more data were provided to the network, there is room for the possibility that it alone may achieve the same performance as SWLDA_WT.

The present study highlights the importance of saving data produced from BCI use. Because BCI systems are expected to be used more often in the future [66], storing data produced from such usage may prove to be valuable, because it can help the system achieve better future performance. Other sources have also indicated the importance of saving data that can be used later for cross-subject training [67], [68]. The same approach could be used and tested with other types of brain signals as well. Large datasets are being opened to the public more and more [69], [70], and we can test whether this approach works for various types of signals using these datasets, a task which is left for future work. Furthermore, there are people who cannot use BCI systems as effectively as other people [71], [72]. These people are considered to have BCI illiteracy, as classifiers are not effectively constructed for these people [73]. Although not dealt with in this manuscript, one could investigate whether such zero-trained networks can help solve the problem of BCI-illiteracy, another task which is also left for future work.

Another point to note about CNN architecture is that our work is not limited only to certain types of CNN architecture. Any CNN would work as well. However, it is recommended to increase the number of kernels or layers optimally in order to reach full performance, because CNN requires sufficient parameters to store various forms of ERP signals. We tested our zero-training approach with other classifiers as well, but ultimately failed to achieve satisfactory performance, as shown in Table. 1. CNN's advantage is that it can classify what it has "seen" in the past. Although ERP signals from different individuals differ, some are actually quite similar. Once a CNN has seen such data, it can then accurately classify other data that do not differ greatly from what it has previously seen.

It is also worth noting that not everyone's data can enhance classification accuracy. We demonstrated what kind of data may be suitable for transfer learning-based zero-training, but some subjects' data may even hinder the performance to a certain extent, as mentioned in this research. Subjects with low mean P300 amplitude tend to influence the network in a bad way. However, besides excluding datasets with

low mean P300 amplitude, accurately selecting a “good” dataset in order to improve performance and to reduce the required number of samples remains a task for future work.

Additionally, the present study demonstrated only an example of reusing a classifier’s output as a true label. It does not clearly show whether or not the increased performance is attributable to certain classifiers’ robustness, or to different features to which nonlinear and linear classifiers react. Although our preliminary analysis revealed no noteworthy difference between features to which the zero- and within-trained networks reacted, such a difference could be investigated further as well.

In summary, this research demonstrated a simple way to achieve BCI without calibration by using data acquired previously from different subjects. This work also investigated how large data combined with a neural network can overcome certain problems in the BCI system, and also provided a hybrid approach to further enhance accuracy by cleverly manipulating them. As BCI systems are expected to be used more and more frequently in the future, this work provides insight into the importance of saving any data acquired from their usage.

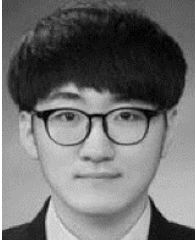
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